Regression Report — Predictive Optimization Inventory Gate

Live end-to-end + unit regression — Deep Clone Cross-Region Migration Orchestrator

Workspace: **vk-west** (`adb-7405612780454429.9.azuredatabricks.net`) • Warehouse `345e7fb63d50405e` • Control tables `po_ctl_t2.migration_meta` • Oct 21, 2026

Result: PASS. All 3 live scenarios behaved exactly as designed, and the full unit suite is green (**70 passed**, incl. 10 PO-specific tests). The gate was exercised against **real** orchestrator modules (`InputResolver` , `PredictiveOptimizationChecker` , `InventoryManager` , `plan_chunks` , Deep Clone, row-count validation) hitting the live Unity Catalog REST API.

1. Test fixtures (created live in vk-west)

Object	PO setting	Tables (rows)
`po_e2e_cat_a` (catalog)	DISABLE	—
↳ `sa_ok`	DISABLE (inherits)	`t_a1` =5, `t_a2` =7
↳ `sa_on`	**ENABLE (schema)**	`t_sch1` =2, `t_sch2` =3
↳ `sa_mixed`	DISABLE	`t_m_ok` =4 (DISABLE), `t_m_on` =4 (**ENABLE table**)
`po_e2e_cat_b` (catalog)	**ENABLE (catalog)**	—
↳ `sb`	ENABLE (inherits)	`t_b1` =6, `t_b2` =6
`po_e2e_cat_c` (catalog)	DISABLE	—
↳ `sc1`	DISABLE (inherits)	`t_c1a` =8, `t_c1b` =9
`po_ctl_t2.po_e2e_clean`	DISABLE	`t_single` =10
`po_e2e_tgt` (catalog)	target	Deep Clone destination

2. Scenario 1 — all selections qualify for failure

CSV: table row (`sa_mixed.t_m_on` , table-level ENABLE), schema row (`sa_on` , schema-level ENABLE), catalog row (`po_e2e_cat_b` , catalog-level ENABLE). **Expected:** 5 resolved, all blocked, 0 queued, step fails.

Source table	Detected level	Outcome
`sa_mixed.t_m_on`	TABLE	PO_BLOCKED
`sa_on.t_sch1`	SCHEMA (sa_on)	PO_BLOCKED
`sa_on.t_sch2`	SCHEMA (sa_on)	PO_BLOCKED
`sb.t_b1`	CATALOG (po_e2e_cat_b)	PO_BLOCKED
`sb.t_b2`	CATALOG (po_e2e_cat_b)	PO_BLOCKED

Observed: `total=5 inserted(QUEUED)=0 po_blocked=5 failed=0` ; inventory step fails. **PASS** — inheritance level correctly identified as TABLE / SCHEMA / CATALOG in each case.

3. Scenario 2 — mixed (qualifying + non-qualifying)

CSV: schema `sa_mixed` (one clean, one table-ENABLE), schema `sa_on` (ENABLE), catalog `po_e2e_cat_b` (ENABLE), table `po_e2e_clean.t_single` (clean). **Expected:** clean tables queued, offenders blocked, step still fails.

Source table	Outcome

`sa_mixed.t_m_ok`	QUEUED
`po_e2e_clean.t_single`	QUEUED
`sa_mixed.t_m_on`	PO_BLOCKED (TABLE)
`sa_on.t_sch1` / `sa_on.t_sch2`	PO_BLOCKED (SCHEMA)
`sb.t_b1` / `sb.t_b2`	PO_BLOCKED (CATALOG)

Observed: `total=7 inserted(QUEUED)=2 po_blocked=5 failed=0`; step fails after recording. **PASS** — confirms the chosen "queue the clean, block offenders, then fail" policy.

4. Scenario 3 — all pass → full Deep Clone + validate

CSV: catalog `po_e2e_cat_c`, schema `sa_ok`, table `po_e2e_clean.t_single` (all PO-disabled). **Expected:** all queued, chunked, Deep Cloned, row counts validated, process completes.

Source → Target	src rows	tgt rows	Validation
`sc1.t_c1a` → `po_e2e_tgt.sc1.t_c1a`	8	8	MATCH
`sc1.t_c1b` → `po_e2e_tgt.sc1.t_c1b`	9	9	MATCH
`sa_ok.t_a1` → `po_e2e_tgt.sa_ok.t_a1`	5	5	MATCH
`sa_ok.t_a2` → `po_e2e_tgt.sa_ok.t_a2`	7	7	MATCH
`po_e2e_clean.t_single` → `po_e2e_tgt.po_e2e_clean.t_single`	10	10	MATCH

Observed: `total=5 inserted(QUEUED)=5 po_blocked=0` → planned **1 chunk (5 tables)** → `CREATE OR REPLACE TABLE … DEEP CLONE …` executed → **5/5 row-count matches**, all marked `COMPLETED` / `VALIDATED`. **PASS** — clean selections flow end-to-end unchanged.

5. Unit regression suite

```
$ python3 -m pytest tests/ -q
....................................................................  [100%]
70 passed, 1 warning in 1.93s
```

- **10** PO-specific tests — flag parsing (ENABLE / DISABLE / INHERIT / missing / API-error fail-closed), parallel `check_selections`, and `InventoryManager` integration.
- **60** pre-existing orchestrator tests — all still green (no regressions).
- Excluded: 3 modules that require the Databricks `dbutils` runtime and 1 notebook with a pre-existing Python-3.9 f-string syntax quirk — unrelated to this change.

6. Coverage summary

Capability	Verified by	Result
Table-level PO detected	S1/S2 (`t_m_on`)	PASS
Schema-level inheritance detected	S1/S2 (`sa_on`)	PASS
Catalog-level inheritance detected	S1/S2 (`po_e2e_cat_b`)	PASS
Clean tables still queue	S2/S3	PASS
Offenders recorded as `PO_BLOCKED` + error_code	S1/S2 control-table rows	PASS

Step fails when any offender present	S1/S2	**PASS**
Parallelism preserved (no sequential pass)	4-thread inventory worker	**PASS**
CSV-driven across catalog/schema/table rows	all 3 scenarios	**PASS**
Deep Clone + row-count validation on clean set	S3	**PASS**
Fail-closed on unreadable PO state	unit test	**PASS**

Regression evidence captured from live runs against vk-west on Oct 21, 2026. Fixtures are isolated under `po_e2e_*` catalogs and `po_ctl_t2` and can be dropped after review.